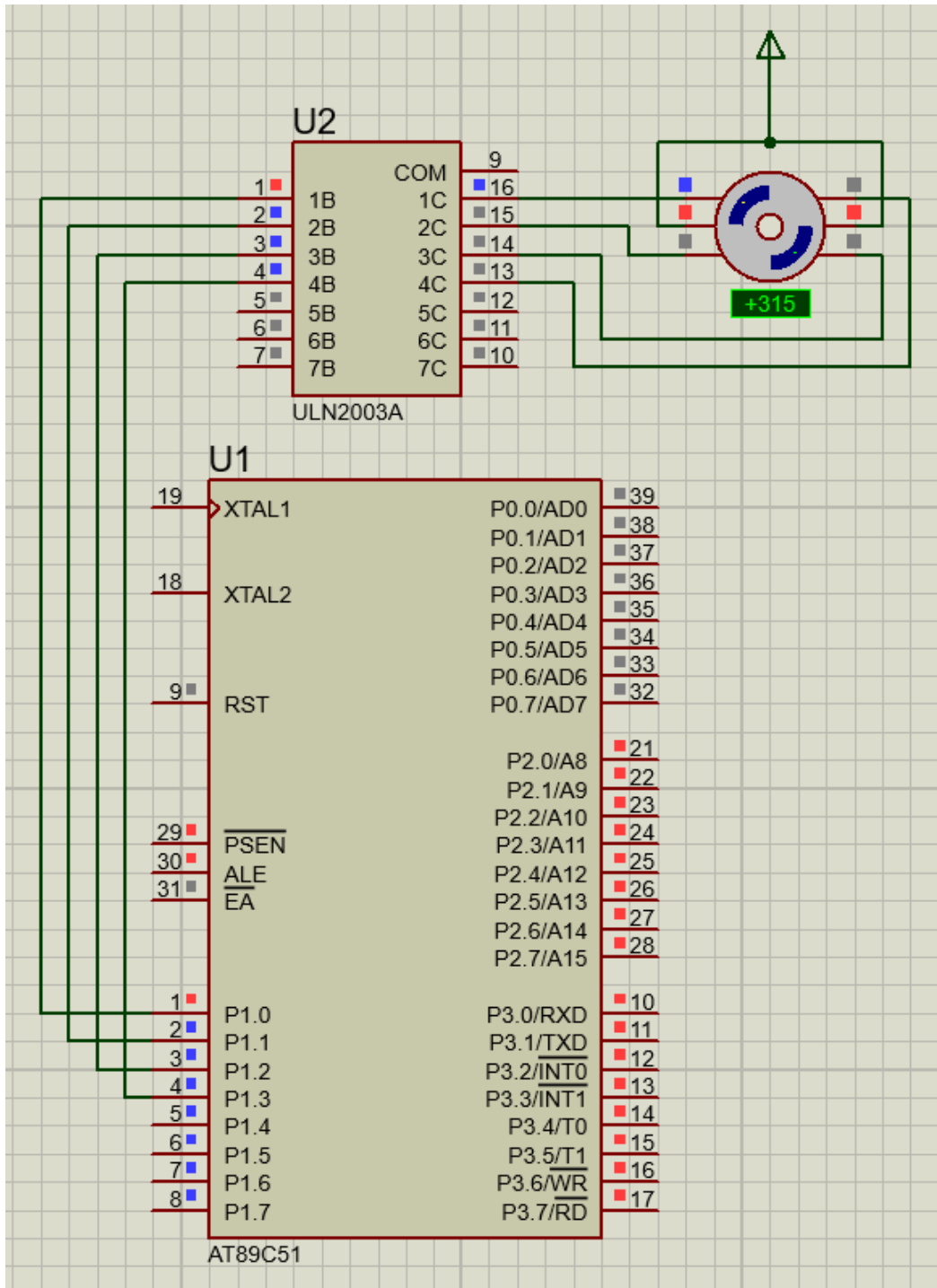
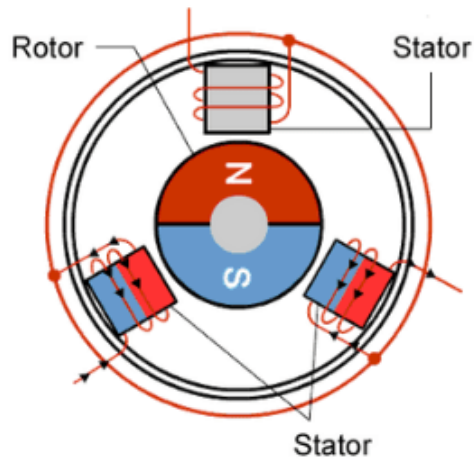


6 Stepper Motor - Rotation

Aim: To interface a stepper motor with the 8051 microcontroller & rotate it.





Apparatus Required:

- 8051 microcontroller
- Stepper Motor
- ULN2003 driver
- Power Supply

Theory:

- A stepper motor is an electro-mechanical device that converts electrical pulses into discrete mechanical movements.
- Unlike DC motors, a stepper motor rotates in fixed angles called steps, which allows precise control of position & speed.
- The rotation of a stepper motor is achieved by energizing its coil in a specific sequence.
- Each sequence step causes the motor shaft to rotate by a small angle.
- By continuously applying these sequences, the motor rotates smoothly.
- Stepper motors can operate in different modes such as full-step & half-step mode.
- In this experiment, a simple stepping sequence is used to rotate the motor continuously.

Drive Modes & Sequences:

1. Wave Drive Mode:

- In this mode, one coil is energized at a time.

- All 4 coils are energized one after another.
- Sequential signals are sent → coils are energized → motor rotates step by step.
- Direction depends on the sequence order.

Step	Winding A	Winding B	Winding C	Winding D
1	1	0	0	0
2	0	1	0	0
3	0	0	1	0
4	0	0	0	1

2. Full Drive Mode:

- Two coils are energized at the same time.
- This mode produces more torque, but power consumption is high.

Step	Winding A	Winding B	Winding C	Winding D
1	1	1	0	0
2	0	1	1	0
3	0	0	1	1
4	1	0	0	1

3. Half Drive Mode:

- 1 and 2 coils are energized alternately.
- At first, one coil is energized, then 2 coils are energized.
- It is a combination (wave + full) drive mode.
- The angular rotation of the motor is increased (higher resolution).

Step	Winding A	Winding B	Winding C	Winding D
1	1	0	0	0
2	1	1	0	0
3	0	1	0	0
4	0	1	1	0
5	0	0	1	0
6	0	0	1	1
7	0	0	0	1
8	1	0	0	1

Program Code:

```

#include <reg51.h>

void delay(int ms) {
    unsigned int i, j;
    for(i = 0; i < ms; i++) {
        for(j = 0; j < 1275; j++);
    }
}

void main() {
    // UNCOMMENT ONLY ONE OF THESE AT A TIME:

    // 1. Wave drive (4 steps)
    int rot_angle[] = {0x08, 0x04, 0x02, 0x01};

    // 2. Full-step (4 steps)
    // int rot_angle[] = {0x0C, 0x06, 0x03, 0x09};

    // 3. Half-step (8 steps)
    // int rot_angle[] = {0x08, 0x0C, 0x04, 0x06, 0x02, 0x03, 0x01, 0x09};

    int i;
    // Calculate the number of items in the array dynamically
    int num_steps = sizeof(rot_angle) / sizeof(rot_angle[0]);

    while(1) {
        // Now the loop automatically adjusts to 4 or 8!
        for(i = 0; i < num_steps; i++) {
            P1 = rot_angle[i];
            delay(100);
        }
    }
}

```

Result: The stepper motor rotated successfully.

Precautions:

- Use Driver IC.

- Proper power supply required.